

FACULTY OF ENGINEERING AND TECHNOLOGY
FINAL ASSESSMENT FOR THE
BACHELOR OF SCIENCE (HONOURS) IN COMPUTER SCIENCE
BSC (HONS) INFORMATION TECHNOLOGY (COMPUTER NETWORKING AND SECURITY)
BACHELOR OF SCIENCE (HONOURS) IN INFORMATION TECHNOLOGY
BACHELOR OF SOFTWARE ENGINEERING (HONS)
& BUSINESS SCHOOL

ACADEMIC SESSION	: MAY 2026 SEMESTER
SUBJECT	: WEB1201 / WEB2014 – WEB FUNDAMENTALS
RELEASE DATE	: 28 MAY 2026 12:00 PM
SUBMISSION DATE	: 10 AUGUST 2026 18:00 MYT
WEIGHTAGE	: 50% (Group Assignment)
MAX MEMBER in 1 GROUP	: 4

ASSIGNMENT QUESTION

WEB1201 / WEB2014: Interactive Multi-Page Web Application Development

1. OVERVIEW & SCENARIO

As the world becomes increasingly digital, businesses and organisations of all sizes rely on professional, functional, and accessible websites to engage with their audiences. Your team has been contracted as a web development consultancy to design and build a complete, multi-page interactive web application for a client organisation of your choice.

In the first part of the group assignment, you submitted a proposal outlining your chosen client, website purpose, and initial design wireframes. Building on that foundation, your team is now required to fully develop and deliver the proposed web application using HTML, CSS, and JavaScript ONLY. The use of any external CSS frameworks (e.g., Bootstrap, Tailwind) or JavaScript libraries/frameworks (e.g., jQuery, React, Vue) is strictly prohibited.

The final deliverable must demonstrate a thorough mastery of core web technologies, dynamic interactivity, accessibility best practices, professional documentation, and individual contribution accountability.

2. REQUIRED WEBSITE IMPLEMENTATION

A. Core Structure, Styling & Theming

- Develop all required HTML pages with well-structured, semantic markup (correct use of <header>, <nav>, <main>, <section>, <footer>, etc.).
- Apply original, cohesive CSS styling: colour palette, typography hierarchy, spacing, and visual theme must be consistent across all pages.
- No inline styles permitted; all CSS must be in external stylesheets.

B. Responsiveness & Accessibility

- The entire website must be fully responsive using CSS media queries (mobile, tablet, desktop viewports).
- All pages must be cross-browser compatible (Chrome, Firefox, Edge).
- Implement basic WCAG 2.1 accessibility principles: appropriate alt attributes on all images, sufficient colour contrast, keyboard-navigable menus, and ARIA labels where necessary.

C. JavaScript Interactivity

Using pure JavaScript (no libraries), implement all of the following:

- Multi-step Registration & Login Simulation: A multi-step registration form (minimum 3 steps with progress indicator) with full client-side validation per step.
- Dynamic Product/Service Catalogue: A filterable and searchable catalogue section with at least 8 items. Users must be able to filter by category, sort by attribute (e.g., name or price), and search by keyword – all implemented in JavaScript without page reload.
- Interactive Feedback / Review Module: A form allowing users to submit a star rating (1–5) and a text review. Submitted reviews must dynamically appear on the page.

D. Individual Portfolio Page (Per Team Member)

Each team member must develop a dedicated personal Portfolio Page within the website. This page must:

- Be accessible via a link on the Team Page.
- Include: personal bio, education background, technical skills with visual indicators (e.g., CSS-only progress bars), featured project highlights, and a simulated downloadable CV button.
- Demonstrate advanced CSS techniques: CSS Grid and Flexbox layout, CSS custom properties (variables), CSS animations or transitions, and a print stylesheet.
- Be individually graded; each member is fully responsible for the design, content, and technical quality of their own page.

E. Team & About Page

- A dedicated Team Page listing all members with photos, roles, and a quantified description of each individual's contributions (tasks completed, pages owned, features developed).

3. REQUIRED DELIVERABLES

#	Deliverable	Details
1	Website Source Files	A single .zip file containing all HTML, CSS, and JavaScript files, neatly organised in a logical folder structure. No broken links; all assets must be included.
2	Group Report (Max 25 pages)	Cover page, table of contents, client background & objectives, design rationale, feature implementation documentation with screenshots, testing results table, challenges & solutions, and individual contribution declaration signed by all members.
3	Video Demo (Max 12 minutes)	Screen-recorded demo uploaded to YouTube (public/unlisted). Must show: responsive layout across 3 viewports, all 4 JavaScript features in action, theme toggle, and each individual portfolio page. Link must be embedded in the report.
4	Individual Reflection (Per Member)	Each member submits a short individual reflection document (500–800 words) describing: their specific contributions, skills learned, challenges faced personally, and suggested improvements. Submitted separately via the LMS.

4. GROUP ASSIGNMENT – TASK BREAKDOWN

The following components are assessed as a team and contribute to the Group Assignment grade. All members are expected to collaborate on planning and peer-reviewing each component.

Task	Description	Component	Marks (Group)
G1	Website Design, Layout & CSS Styling Overall visual design, colour theme, CSS structure, cross-page consistency, and creative presentation.	Website Dev	10 marks
G2	Responsiveness & Accessibility Full responsiveness using media queries, cross-browser testing, and WCAG 2.1 compliance (alt text, contrast, ARIA labels).	Website Dev	10 marks
G3	JavaScript Features (Collective Integration) Integration and overall functionality of: multi-step registration/login, catalogue filter & search, review module, and theme toggle. Evaluated as a cohesive feature set.	Website Dev	15 marks
G4	Team & About Page Quality and completeness of the Team page, member roles, contribution descriptions, and navigation to individual portfolio pages.	Website Dev	5 marks
G5	Group Report Professional report covering client background, design rationale, implementation documentation, testing results table, challenges & solutions, and signed contribution declaration.	Report	15 marks
G6	Video Demonstration Screen-recorded YouTube video clearly demonstrating all responsive viewports, all JavaScript features, and individual portfolio pages. Max 12 minutes, link in report.	Video	10 marks
GROUP SUBTOTAL			65 marks

5. INDIVIDUAL ASSIGNMENT – TASK BREAKDOWN

The following components are assessed per individual student. Each member is solely responsible for their own deliverables. Individual marks are NOT shared among the group.

Task	Description	Component	Marks (Individual)
I1	Individual Portfolio Page – Content & Design Personal bio, education, technical skills with visual CSS indicators, project highlights, and simulated CV download. Quality, completeness, and relevance of content.	Website Dev	10 marks
I2	Individual Portfolio Page – Advanced CSS Techniques Evidence of CSS Grid and Flexbox layout, CSS custom properties (variables), CSS animations/transitions, and a functional print stylesheet.	Website Dev	10 marks
I3	Individual Reflection Document 500–800 word reflection discussing: personal tasks completed, skills developed, challenges faced, and recommended improvements. Quality of critical self-evaluation.	Reflection	10 marks
I4	Code Ownership & Quality Well-structured, readable, and commented code for the sections the student is responsible for. Marks awarded based on code quality, logical organisation, and inline documentation.	Website Dev	5 marks
INDIVIDUAL SUBTOTAL			35 marks

GRAND TOTAL (Group + Individual)	100 marks
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6. GROUP ASSIGNMENT – MARKING RUBRIC

G1 – Website Design, Layout & CSS Styling (10 Marks)				
Criteria	Excellent (9–10)	Good (7–8)	Satisfactory (5–6)	Needs Improvement (0–4)
Visual Design & Theme	Exceptional visual appeal; creative, polished theme highly suited to the client.	Strong design; cohesive and appropriate theme with minor inconsistencies.	Adequate design; theme somewhat relevant but lacks refinement.	Poor design; inconsistent or irrelevant theme; unattractive layout.
CSS Structure & Quality	Fully external CSS; well-organised stylesheets; advanced selectors and properties used creatively.	External CSS; mostly organised; good use of CSS properties.	Mostly external CSS; some inline styles; adequate organisation.	Heavy use of inline styles; disorganised; basic or incorrect CSS.
Typography & Colour	Excellent hierarchy; consistent and professional typography and colour palette.	Clear hierarchy; consistent colours with minor deviations.	Some hierarchy; colour choices inconsistent or clashing.	No clear hierarchy; poor or mismatched colour choices.

G2 – Responsiveness & Accessibility (10 Marks)				
Criteria	Excellent (9–10)	Good (7–8)	Satisfactory (5–6)	Needs Improvement (0–4)
Responsiveness (Media Queries)	Seamless adaptation across mobile, tablet, and desktop; all elements reflow correctly.	Responsive on all viewports with minor layout issues on one size.	Partially responsive; some breakpoints missing or broken layout.	Little or no responsiveness; layout broken on smaller screens.
Cross-Browser Compatibility	Fully tested and functions identically on Chrome, Firefox, and Edge.	Works on all three browsers with very minor visual differences.	Works on two browsers; visible issues on one.	Only tested/works on one browser; significant issues on others.
Accessibility (WCAG 2.1)	Full alt text, sufficient contrast, keyboard-navigable menu, ARIA labels; meets WCAG 2.1 Level AA.	Most accessibility criteria met; minor omissions.	Some effort; alt text partially present; contrast issues.	Minimal or no accessibility consideration.

G3 – JavaScript Features – Collective Integration (15 Marks)				
Feature	Excellent (13–15)	Good (10–12)	Satisfactory (6–9)	Needs Improvement (0–5)
Multi-Step Registration & Login	Flawless multi-step form with progress indicator; login with greeting.	All features present; minor validation gap or UI issue.	Form partially complete; validation inconsistent; login simulated	Single-page form only; no validation; login not functional.
Catalogue Filter, Sort & Search	All 8+ items; filter by category, sort by attribute, and keyword search all work dynamically without reload.	All functions present; minor bug in one (e.g., sort not resetting).	Two of three functions work; catalogue items fewer than required.	Only one function works or catalogue is static HTML.
Review Module	Star rating + text review; real-time character count;	All features present; persistence works with minor UI glitch.	Reviews display but do not persist; character count missing.	Form exists but reviews do not appear or are not functional.
Theme Toggle	Light/dark toggle applies across all pages;	Toggle works on most pages; persistence present.	Toggle works on one page only; no persistence.	Toggle absent or non-functional.

G4 – Team & About Page (5 Marks)				
Criteria	Excellent (5)	Good (4)	Satisfactory (3)	Needs Improvement (0–2)
Completeness & Design	All members listed; photos, roles, and quantified contributions; professional design consistent with site theme.	All members listed with photos and roles; minor contribution detail missing.	Most members listed; limited contribution detail; design inconsistent.	Incomplete page; missing members or photos; poor design.
Links to Portfolio Pages	All portfolio page links functional and correctly labelled.	All links present; one broken link.	Some links working; others missing.	Links absent or all broken.

G5 – Group Report (15 Marks)				
Criteria	Excellent (13–15)	Good (10–12)	Satisfactory (6–9)	Needs Improvement (0–5)
Client Background & Design Rationale	Clear, well-researched client profile;	Good client overview; design	Client described briefly; design	Client background vague; no design justification.

	compelling and well-justified design decisions aligned to client needs.	rationale mostly justified.	rationale superficial.	
Implementation & Testing Documentation	Thorough documentation of all features with clear screenshots; complete testing table with expected and actual results for all functionalities.	Most features documented with screenshots; testing table mostly complete.	Some features documented; testing table incomplete or screenshots missing.	Minimal documentation; no testing evidence.
Challenges & Solutions	Comprehensive, specific challenges documented with effective, clearly explained solutions and lessons learned.	Good coverage of challenges with solutions described.	Some challenges mentioned; solutions vague.	No meaningful challenges described.
Report Quality & Formatting	Professional layout; consistent formatting; no grammar/spelling errors; signed contribution declaration included.	Well-formatted with minor errors; declaration present.	Adequate formatting; several errors; declaration present.	Poor formatting; many errors; declaration missing.

G6 – Video Demonstration (10 Marks)				
Criteria	Excellent (9–10)	Good (7–8)	Satisfactory (5–6)	Needs Improvement (0–4)
Feature Coverage & Clarity	All 4 JS features and all portfolio pages clearly demonstrated; narration or captions present.	All features shown; one portfolio page briefly covered; mostly clear.	Most features shown; some functionalities skipped or unclear.	Few features demonstrated; video unclear or incomplete.
Responsive Demo	Clearly shows layout at all 3 viewports; uses browser dev tools or actual devices.	Shows 2 viewports clearly; third briefly or implied.	Only 1 viewport shown; responsiveness demo incomplete.	Responsive demo absent.
Production Quality & Duration	Professional quality; within 12 minutes; link functional and public.	Good quality; slightly over/under time; link works.	Acceptable quality; over time or link has minor access issue.	Poor quality; link broken or video private.

7. INDIVIDUAL ASSIGNMENT – MARKING RUBRIC

I1 – Portfolio Page – Content & Design (10 Marks)				
Criteria	Excellent (9–10)	Good (7–8)	Satisfactory (5–6)	Needs Improvement (0–4)
Content Completeness	All sections present: bio, education, skills, projects, and simulated CV download.	All sections present; one section lacks detail.	Most sections present; two or more superficial.	Several sections missing; very sparse content.
Visual Design & Professionalism	Polished, professional design; clearly portfolio-grade; consistent with site theme.	Professional appearance with minor inconsistencies.	Adequate design; does not feel professional.	Poor design; clearly minimal effort.
Skill Indicators (CSS Progress Bars)	Creative, well-implemented CSS-only skill bars; visually clear and accurately labelled.	Skill bars present and functional; minor styling issues.	Basic skill representation; bars not CSS-only or inaccurate.	Skill indicators absent or non-functional.

I2 – Portfolio Page – Advanced CSS Techniques (10 Marks)				
Criteria	Excellent (9–10)	Good (7–8)	Satisfactory (5–6)	Needs Improvement (0–4)
CSS Grid & Flexbox	Both Grid and Flexbox used meaningfully and correctly; layout is structurally elegant.	Both used; one applied less effectively.	Only one layout technique used, or used incorrectly.	Neither Grid nor Flexbox used.
CSS Custom Properties (Variables)	Consistent use of :root variables for colours, fonts, and spacing throughout the page.	Variables used for colours; limited scope.	One or two variables defined; not systematically used.	No CSS custom properties used.
Animations & Transitions	Smooth, purposeful CSS animations or transitions enhance UX (e.g., hover, entrance, scroll effects).	Transitions present; animations somewhat generic.	One basic transition; no animations.	No animations or transitions.

Print Stylesheet	Print stylesheet hides navigation, adjusts layout for print, and is clearly linked with media='print'.	Print stylesheet present; minor formatting issues in print view.	Basic print stylesheet; navigation not hidden.	No print stylesheet.
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I3 – Individual Reflection Document (10 Marks)				
Criteria	Excellent (9–10)	Good (7–8)	Satisfactory (5–6)	Needs Improvement (0–4)
Contribution Description	Specific, clearly articulated account of personal tasks; features developed named and evidenced.	Good description of contributions; mostly specific.	Contributions described at high level; lacking specifics.	Vague; no clear ownership of tasks.
Skills & Learning Outcomes	Insightful reflection on specific technical and soft skills gained; connects learning to course outcomes.	Reflects on skills gained; connection to course outcomes adequate.	Skills mentioned briefly; superficial reflection.	No meaningful learning reflection.
Critical Evaluation & Improvements	Thoughtful critique of personal and team performance; specific, actionable improvement suggestions.	Some critique offered; improvements somewhat vague.	Minimal critique; improvements generic.	No evaluation or improvement suggestions.
Writing Quality & Word Count	500–800 words; professional, error-free writing; well-structured paragraphs.	Within word count; mostly clear writing with minor errors.	Slightly outside word count; several grammatical errors.	Significantly under/over word count; very poor writing quality.

I4 – Code Ownership & Quality (5 Marks)				
Criteria	Excellent (5)	Good (4)	Satisfactory (3)	Needs Improvement (0–2)
Readability & Commenting	Code is clearly commented; meaningful variable/function names; logically	Mostly commented and readable; minor naming issues.	Some comments; code readable but not well-organised.	No comments; poor naming; difficult to follow.

Code Organisation & File Structure	structured and easy to follow.			
	Separate CSS and JS files; logical folder structure; no unused or duplicate code.	Good structure; minor duplication or misplaced files.	Mostly separated; some inline code; modest organisation.	Mixed/inline code throughout; disorganised files.

8. MARKS SUMMARY

Task	Component	Type	Marks
G1	Website Design, Layout & CSS Styling	Group	10
G2	Responsiveness & Accessibility	Group	10
G3	JavaScript Features – Collective Integration	Group	15
G4	Team & About Page	Group	5
G5	Group Report	Group	15
G6	Video Demonstration	Group	10
Group Subtotal			65
I1	Portfolio Page – Content & Design	Individual	10
I2	Portfolio Page – Advanced CSS Techniques	Individual	10
I3	Individual Reflection Document	Individual	10
I4	Code Ownership & Quality	Individual	5
Individual Subtotal			35
GRAND TOTAL			100

9. SUBMISSION GUIDELINES & ACADEMIC INTEGRITY

Submission Instructions

- Upload all deliverables via the elearn by 18:00 MYT on 10 August 2026.
- Group submission: one member submits the zipped website files, group report (PDF), and YouTube video link.
- Individual submission: each member submits their reflection document separately via the designated eLearn.
- Label all files clearly: [StudentID]_[Name]_Reflection.pdf and [GroupName]_WEB1201_May2026.zip.

Late Submission Policy

- Submissions received within 24 hours after the deadline will incur a 20% deduction of the total marks obtained.
- Submissions more than 24 hours late will receive zero marks for the affected component.

Academic Integrity

- All submitted work must be original. Any use of AI-generated code submitted as your own, plagiarism, or code sharing between groups will be treated as academic misconduct.
- The signed Individual Contribution Declaration in the group report is mandatory and confirms each member's honest account of their contribution.
- Lecturers reserve the right to conduct an interview to verify individual understanding and contribution.